



SESSION 16

Mastering Vectors

Geometric & Algebraic Intuition · Dot Products · Vector Projections ·
Vector Norms

Geometric vs Algebraic Representation

Basic Vector Operations

Dot Product (Algebraic & Geometric)

Vector Projections

Vector Norms (L_1 , L_2 , L_∞)

Linear Combinations & Span

Linear Independence

Basis Vectors

Session Notes — Complete Reference

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Table of Contents

01 Geometric vs Algebraic Representation

02 Basic Vector Operations

03 Dot Product (Algebraic & Geometric)

04 Vector Projections

05 Vector Norms (L_1 , L_2 , L_∞)

06 Linear Combinations & Span

07 Linear Independence

08 Basis Vectors

09 Quick Reference Card

01 • Geometric vs Algebraic Representation

Geometric: A vector is an arrow in space with **magnitude** (length) and **direction**.

Algebraic: An ordered list of numbers represented as a column vector $\mathbf{v} = [v_1, v_2, \dots, v_n]^T$.

02 • Basic Vector Operations

- **Vector Addition:** $\mathbf{u} + \mathbf{v} = [u_1 + v_1, u_2 + v_2]^T$. Geometrically represented by head-to-tail addition.
- **Scalar Multiplication:** $c\mathbf{v} = [cv_1, cv_2]^T$. Scales the vector's length without changing its line of action.

03 · Dot Product (Algebraic & Geometric)

DOT PRODUCT FORMULAS

Algebraic: $\mathbf{u} \cdot \mathbf{v} = \sum u_i \cdot v_i$ Geometric: $\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\| \cdot \|\mathbf{v}\| \cdot \cos(\theta)$

If $\mathbf{u} \cdot \mathbf{v} = 0$, the vectors are **orthogonal** (perpendicular).

04 · Vector Projections

The projection of vector $\mathbf{u}$ onto vector $\mathbf{v}$ yields the shadow of $\mathbf{u}$ along the direction of $\mathbf{v}$:

VECTOR PROJECTION FORMULA

$$\text{proj}_v(u) = \left[(\mathbf{u} \cdot \mathbf{v}) / \|\mathbf{v}\|^2 \right] \cdot \mathbf{v}$$

05 · Vector Norms (L1, L2, L-infinity)

Norms measure the length/magnitude of vectors:

- **L_1 Norm (Manhattan):** $\|\mathbf{x}\|_1 = \sum |x_i|$. Used in Lasso regularization.
- **L_2 Norm (Euclidean):** $\|\mathbf{x}\|_2 = \sqrt{\sum x_i^2}$. Used in Ridge regularization.
- **L_∞ Norm (Chebyshev):** $\|\mathbf{x}\|_\infty = \max |x_i|$.

06 • Linear Combinations & Span

A **linear combination** of vectors is a sum of scaled vectors: $\mathbf{w} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2$. The **span** of a set of vectors is the set of all possible linear combinations that can be formed from them.

07 • Linear Independence

A set of vectors is **linearly independent** if no vector in the set can be written as a linear combination of the others. None of them are redundant.

08 • Basis Vectors

A **basis** for a vector space is a set of linearly independent vectors that span the entire space. Standard basis in $\mathbb{R}^2$ is $\mathbf{i} = [1, 0]^T$ and $\mathbf{j} = [0, 1]^T$.

09 · Quick Reference Card



Vectors Cheat Sheet

- **Orthogonality:** $\mathbf{u} \cdot \mathbf{v} = 0$.
- **Euclidean length (L_2 norm):** $\sqrt{\mathbf{u} \cdot \mathbf{u}}$.
- **Unit Vector:** Vector with norm = 1, computed as $\mathbf{v} / \|\mathbf{v}\|$.